Std9

chapter4

work sheet

MULTIPLICATION IDENTITIES

1. **Mentally find the products below**

   i) 41 x 51 ii) 32 X 62 iii) 91/2×51/2 iv)7.5X3.5 v)91/4×3¼

2) The product of two numbers is 936 and their sum is 61 what is the product of the numbers next to each?

3) The product of two odd numbers is 861 and their sum is 60 what is the product of the odd numbers next to each?

4) The product of two even numbers is 1512 and their sum is 78 what is the product of the even numbers next to each?

5) Calculate each of the following for several numbers and guess a generalisation then prove it using algebra

i) The remainder got on division by 4,the product of two numbers one of which leaves remainder 1 on division by 4 and other ,remainder 3

ii) The reminder got on division by 5, the product of two numbers one of which leaves remainder 1 on division by 5 and other reminder 2

iii) The remainder got on division by 7, the product of two numbers one of which leaves remainder 1 on division by 7 and the other remainder 2

iv) The difference of the product of the two numbers at the ends and the two in the middle of 4 consecutive natural numbers

v) What is the difference between the product of the two end numbers and the square of the middle number of five consecutive natural numbers?

vi) What is the difference between the product of the first and last numbers and the square of the middle number among five consecutive odd numbers?

vii) What is the difference between the product of the two end numbers and the product of the two middle numbers among four consecutive even numbers?

6) The perimeter of a rectangle is 56 cm and the area is160 square centimetres find the area of the rectangle with the each side 3 cm shorter.

7) When both the length and the breadth of a rectangular plot are reduced by 2 metres each, its area becomes 660 square metres. When both the length and the breadth are increased by 2 metres each, its area becomes 884 square metres.

i) What is the area of the original plot?

Ii) What is the perimeter of the original plot?

iii) What are the length and the breadth of the original plot?

8) When both the length and the breadth of a rectangular plot are reduced by 1 metre each, its area becomes 575 square

metres. When both the length and the breadth are increased by 1 metre each, its area becomes 675 square metres.

i) What is the area of the original plot?

ii) What is the perimeter of the original plot?

iii) What are the length and the breadth of the original plot?

9) When each of two numbers are increased by 2 the product becomes 899 and when it is decreased by 2 the product become 675

i) what is the product of the numbers

ii) what is their sum

iii) what are the numbers

10) The product of two odd numbers just after each of two odd numbers is 399 and the product of the odd numbers just before each 255 what are the numbers

11) The product of two even numbers just after each of two even numbers is 840 and the product of the even numbers this before each is 624 what are the numbers?

12) The product of two numbers is 384 and their difference is 8.

i) what is the product of the larger number increased by 1 and the smaller number decreased by 1?

ii) what is the product of the larger number decreased by one and the smaller number increased by one?

13) The product of two numbers is 315 and difference is 4.

i) what is the product of the larger number increased by 2 and the smaller number decreased by 2

ii) what is the product of the larger number decreased by 2 and the smaller number increased by 2

14) The product of the larger of two numbers increased by 1 and the smaller decreased by one is 775. The product of larger decreased by one and the smaller increased by one is 783

i) what is the product of the numbers themselves ?

ii) what is their difference?

iii) what are the numbers?

15) The product of two numbers is 5 more than the product of the larger of the number decreased by 1 and the smaller increased by one how much is the decrease in the product if the larger is increased by 1 and smaller is decreased by one?

16) If the length of a rectangle is increased by 3 m and breadth decreased by 2 m its area would increase by 2 square metres. If the length is decreased by 2 m and breadth is increased by 3 metres the area would increase by 30 square metres calculate length and breadth of the rectangle?

SEEMA SUGATHAN

H. S .T (Maths)